Tanmay Sharma

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Education

Northeastern University **September 2024 – Present**
Master of Science in Cybersecurity - 3.70/4.00 *Boston, MA*

Coursework: Network security, Linux administration, Binary exploitation, Linux kernel security, ARMv8-A, x86, Security risk management and assessment

Technical Skills

Security Ops: Vulnerability Enumeration and Analysis, Secure Code Reviews, Active Directory, Linux Administration
Security Tools: Burp Suite, Metasploit Framework, Nessus, Wireshark, Nmap, Binary Ninja
Programming: C, Python, SQL, Bash, PowerShell, ROP, Socket Programming
Protocols & Standards: OWASP Top 10, CWE, MITRE ATT&CK, HTTP/HTTPS, SSL/TLS, OAuth, TCP/IP
Certifications: CompTIA Security+ , PNPT by TCM Security (In Progress)

Experience

Security Consultant Intern **July 2026 – Present**
Scropion Labs, K logix *Brookline, MA*

- Collaborated on reverse engineering the EdgeTPU driver on Pixel devices, analyzing its interaction with firmware, which enhanced the lab's understanding of hardware security layers.
- Analyzed FirmXRay software by OSUSecLab, focusing on baseaddresssolver.java module, to identify how base addresses are resolved and features extracted, improving vulnerability assessment capabilities in embedded firmware.

Application Security Intern **January 2026 – April 2026**
Artelus.ai *Boston, MA*

- Part of the penetration testing team, followed OWASP Web Security Testing Guidelines, performed reconnaissance, source code review, and exploitation across multiple sprints using Scrum methodology.
- Documented and triaged identified vulnerabilities with actionable remediation guidance, contributing findings to a broader regulatory evidence packet supporting U.S. and EU market entry.

Graduate Teaching Assistant **January 2026 – April 2026**
Khoury School of Computer Science, NU *Boston, MA*

- Conducted weekly lab sessions for a class of 59 students on network security, CTF Boxes, system security, Docker, and Linux administration topics etc while providing hands-on guidance and real-time debugging support.
- Assisting in development of supplementary course materials including lab documentation, Docker environment setup guides, etc while also grading hw assignments. Did QA and testing for Viceroy Decree's Sentinel CTF 2026.

Research Volunteer **March 2025 – August 2025**
CyberspACe securiTy and forensIcs lab (CactiLab) *Boston, MA*

- Collaborated on reverse engineering the EdgeTPU driver on Pixel devices, analyzing its interaction with firmware, which enhanced the lab's understanding of hardware security layers.
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Projects

Kernel Sanders | *C, ARMv8-A, Linux, Shell, Python, TCP Sockets* **February 2026 – April 2026**

- Exploited a vulnerable AArch64 network service to deliver a 544-byte position-independent shellcode beachhead, then escalated to root by directly zeroing credential fields in the kernel's task struct via a custom-mapped RWX exploit.
- Deployed a 7-module Linux rootkit that hooks 15 kernel functions for process and file hiding, symlink blocking, and log sanitization, with a covert C2 channel tunneled through signal 62 supporting 10 operator commands including live shellcode injection.

Active Directory Lab | *AD, LLMNR, Bloodhound, Kerberoasting* **November 2025**

- Designed and deployed a complete Active Directory pentesting environment with Windows Server 2022 Domain Controller and 2 Windows 10 Enterprise clients, configuring certificate services, domain forests, and realistic user scenarios to simulate enterprise infrastructure vulnerabilities
- Executed comprehensive security assessment covering 15+ attack vectors including LLMNR poisoning, SMB relay, Kerberoasting, and Golden Ticket attacks, documenting exploitation methodologies and defense strategies through systematic enumeration with BloodHound, ldapdomaindump, and PingCastle

Secure Instant Messaging System | *Python, SRP, ECDH, TCP Sockets* **January 2025 – April 2025**

- Built a secure terminal-based chat system using SRP, ECDH, and AES-GCM for confidential, authenticated real-time communication (CY6740 - Network Security).
- Developed socket-based architecture with encrypted commands, rate limiting, and user listing, cutting abuse vectors by 80% and boosting resilience against attacks.